Numbers: Introduction

COMP0147

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Term 1 – 2026/27



Introduction

- Sets are a fundamental structure of mathematics.

  In particular, number sets are of utmost importance.

- Here we will briefly cover some important sets of numbers:
  - the natural numbers $\mathbb{N}$,
  - the integers $\mathbb{Z}$,
  - the rational numbers $\mathbb{Q}$,
  - the irrational numbers $\mathbb{I}$, and
  - the real numbers $\mathbb{R}$.

The Natural Numbers

- The set of natural numbers is the set

$$\mathbb{N} = \{0, 1, 2, 3 \ldots\}.$$

Some authors do not consider the number 0 (zero) as a natural number, defining $\mathbb{N} = \{1, 2, 3, \ldots\}$.

In this course we will consider 0 as a natural number.

- The set of natural numbers $\mathbb{N}$ can be defined through two "self-evident observations":

  N_1: 0 (zero) is a natural number, and

  N_2: every natural number has a successor.

The Natural Numbers

- Rewriting these "self-evident observations" in a more formal way, we obtain the following axioms for the naturals:

  N_1': $0 \in \mathbb{N}$, and

  N_2': if $k \in \mathbb{N}$, then $s(k) \in \mathbb{N}$,

  where $s(\cdot)$ is the successor function: $s(k) = k + 1$.

- Examples:

  1. $0 \in \mathbb{N}$, because of (N_1').
  2. $s(0) \in \mathbb{N}$, because of (N_2'). Notation: $s(0) = 1$.
  3. $s(s(s(s(s(0))))) = 5 \in \mathbb{N}$.

- To obtain the natural number n, apply (N_2') n times.

Natural Numbers in Decimal Representation

- Natural numbers can be decomposed into terms written as powers of 10.
- Example 1:

$$723 \quad =$$

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$$\begin{aligned} 723 &= 7 \cdot 10^2 + 2 \cdot 10^1 + 3 \cdot 10^0 \\ &= 700 + 20 + 3 \\ &= 723 \end{aligned}$$

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- Example 2:

$$\begin{aligned} 8\,007 &= 8 \cdot 10^3 + 0 \cdot 10^2 + 0 \cdot 10^1 + 7 \cdot 10^0 \\ &= 8\,000 + 7 \\ &= 8\,007 \end{aligned}$$

◁

Natural Numbers in Binary Representation

- There is nothing special about the choice of powers of 10 to decompose natural numbers.

  We can represent natural numbers in powers of 2, for example.

- Example 3:

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Natural Numbers in Binary Representation

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$$\begin{aligned} 1010_2 &= 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 \\ &= 8 + 2 \\ &= 10 \end{aligned}$$

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- Example 4:

$$110111_2 \quad =$$

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- Example 4:

$$\begin{aligned} 110111_2 &= 1 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 \\ &= 32 + 16 + 4 + 2 + 1 \\ &= 55 \end{aligned}$$

Natural Numbers in Binary Representation

- A quick method to represent a natural number in binary:

Binary digit	1	1	0	1	1	1	
Position in the number	5	4	3	2	1	0	
Decimal equivalent: 2^{position}	32	16	8	4	2	1	
Contribution of each digit	32	16	0	4	2	1	55

- Inverse operation:

$$77 \xrightarrow[\text{remainder}=1]{div2} 38 \xrightarrow[\text{remainder}=0]{div2} 19 \xrightarrow[\text{remainder}=1]{div2} 9 \xrightarrow[\text{remainder}=1]{div2} 4 \xrightarrow[\text{remainder}=0]{div2} 2 \xrightarrow[\text{remainder}=0]{div2} 1 \xrightarrow[\text{remainder}=1]{div2} 0$$

The bits are produced from least significant to most significant, so the binary representation is obtained by reversing the order of the bits produced:

$$77 \quad = \quad 1001101.$$

The Integers

- The set of integers is the set

$$\mathbb{Z} = \{-3, -2, -1, 0, 1, 2, 3 \ldots\}.$$

The set

$$\mathbb{Z}^+ = \{1, 2, 3, 4, 5, \ldots\}$$

is the set of positive integers.

- The set of integers $\mathbb{Z}$ can be defined as the set of all natural numbers and their negatives:

$$\mathbb{Z} = \{x \mid x \in \mathbb{N} \quad \text{or} \quad -x \in \mathbb{N}\}.$$

The Real Numbers

- Another important set is the set of real numbers $\mathbb{R}$.

- Examples:

  1. $\pi = 3.14159265359\ldots$
  2. $e = 2.71828182846\ldots$
  3. $\sqrt{2} = 1.41421356237\ldots$
  4. $-\ln 2 = -0.6931471805\ldots$
  5. 2
  6. -4.5

- A real number can be defined as an infinite sum:

$$d_k \quad d_{k-1} \quad \cdots \quad d_1 \quad d_0 \quad . \quad d_{-1} \quad d_{-2} \quad d_{-3} \quad \cdots \quad = \quad \sum_{i=-\infty}^{k} d_i \cdot 10^i$$

The Real Numbers

- If a real number ends in an infinite sequence of 9s, replace all 9s with 0s and increment the digit before the first 9.

$$0.99999999\cdots \quad = \quad 1.00000000\cdots \quad = \quad 1.$$

Intuitively, this rule makes sense because:

$$1 \quad = \quad 3 \cdot \frac{1}{3} \quad = \quad 3 \cdot 0.3333333\ldots \quad = \quad 0.9999999\ldots$$

More formally, we can verify that:

$$\begin{aligned} 0.99999999\ldots &= \frac{9}{10} + \frac{9}{10^2} + \frac{9}{10^3} + \frac{9}{10^4} + \cdots \\ &= \frac{9/10}{1 - 1/10} && \text{(inf. geom. series sum)} \\ &= \frac{9/10}{9/10} \\ &= 1 \end{aligned}$$

Real Numbers in Binary Representation

- Examples:
  - $1010 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 = 8 + 2 = 10$
  - $0.1 = 1 \cdot 2^{-1} = 1/2$
  - $0.01 = 1 \cdot 2^{-2} = 1/4$
  - $0.11 = 1 \cdot 2^{-1} + 1 \cdot 2^{-2} = 3/4$
- Definition 1. A real number is an infinite sum:

$$b_k b_{k-1} \ldots b_1 b_0 \, . \, b_{-1} b_{-2} b_{-3} \ldots = \sum_{i=-\infty}^{k} b_i 2^i$$

- Warning:
  If a real ends in an infinite sequence of 1s, replace all the *1*s with *0*s, and increment the bit before the first *1*, from *0* to *1*.
  $0.11111111\cdots = 1.00000000\cdots = 1$

The Rational Numbers

- The next set of interest is the set of rational numbers $\mathbb{Q}$.
- A rational number is a real number x such that there exist $p, q \in \mathbb{Z}$, with $q \neq 0$, such that

$$x = \frac{p}{q}.$$

- Examples:

1. $\frac{1}{2}$
2. $\frac{-1}{2} = \frac{1}{-2} = -\frac{1}{2}$
3. $\frac{7\,327\,569\,219\,559}{6\,576\,576\,329\,756}$

The Rational Numbers

- Example 5:
  - $10/3$ is rational?

    Yes. Quotient of integers.
  - 0.281 is rational?

    Yes. Decimal notation number representing $281/1000$.
  - Is every number represented on a traditional calculator rational?

    Yes. The calculator display is finite and for that reason all represented numbers are rational.
  - $0.121212\ldots$ is rational? Yes. Let $x = 0.121212\ldots$ and $100x = 12.121212\ldots$.

$$\begin{aligned} 100x - x &= 12.121212\ldots - 0.121212\ldots \\ 99x &= 12 \\ x &= 12/99 \end{aligned}$$

The Rational Numbers

- Theorem 1. A real number x is rational if, and only if, its decimal representation is periodic.

  Examples:

  1. $1/5 \quad = \quad 0.2\,0000\,0000\,0000\,\ldots$
  2. $1/7 \quad = \quad 0.142857\,142857\,142857\,\ldots$

The Rational Numbers

- We have already seen that a rational number is a real number x such that there exist $p, q \in \mathbb{Z}$, with $q \neq 0$, such that

$$x = \frac{p}{q}.$$

- Note that the condition for two rational numbers to be equivalent, or equal, is the following:

$$\frac{p}{q} = \frac{r}{s} \quad \text{if, and only if,} \quad ps = qr.$$

- We can always choose as the representation of a rational number the simplified representation:

$$\text{a fraction where } \gcd(p, q) = 1.$$

The Irrational Numbers

- The following result shows that not every number is rational, that is, there exist irrational numbers (whose set is $\mathbb{I}$).

- Theorem 2. $\sqrt{2}$ is not rational.

  Proof. By contradiction.

  Assume the opposite of what we want to prove, i.e., that $\sqrt{2}$ is rational.

  In this case, we know there exist numbers $p, q \in \mathbb{Z}$, with $\gcd(p, q) = 1$, such that $\sqrt{2} = p/q$.

  Squaring both sides of the above equation, we get $2 = p^2/q^2$, that is, $p^2 = 2q^2$.

  Note that $2q^2$ is even, therefore by the equality above p^2 must also be even. This implies that p must be even.

Irrational numbers

- Proof (Continuation).

  Now, since p is even, there exists some $s \in \mathbb{Z}$ such that $p = 2s$. This implies that $2q^2 = p^2 = (2s)^2 = 4s^2$, which results in $q^2 = 2s^2$. Note that then q^2 is even, therefore q must be even.

  But if both p and q are even, this contradicts the assumption that $\gcd(p, q) = 1$: we have reached a contradiction.

  Conclusion: there do not exist $p, q \in \mathbb{Z}$, with $q \neq 0$ and $\gcd(p, q) = 1$, such that $\sqrt{2} = p/q$, therefore $\sqrt{2}$ is not rational.

□

Rational versus irrational

Some facts about rationals and irrationals:

1. The set of real numbers is the union of rationals and irrationals: $\mathbb{R} = \mathbb{Q} \cup \mathbb{I}$.
2. Between two rational numbers there always exists an irrational number.
3. Between two irrational numbers there always exists a rational number.
4. There exist irrational numbers, for example $\sqrt{2}$.
5. The number π is not a root of any polynomial with coefficients in $\mathbb{Q}$. This makes π a transcendental number. Those that are roots of such polynomials are called algebraic numbers.
6. The sum of two rational numbers is rational.
7. The sum of a rational and an irrational number is irrational.
8. The sum of two irrationals is tricky:
   it is not known whether $\pi + e \in \mathbb{Q}$ or $\pi + e \notin \mathbb{Q}$.
9. There exist irrational numbers x and y such that x^y is rational.

Rational versus irrational

Spoiler alert!

The "size" of the set of natural numbers and that of rationals is the same, but the "size" of the set of real numbers is larger.

(We will go deeper into this later in the course...)